

In other words, the formula is obtained by deleting the last term in the Schwarz-Christoffel integral (5).

Proof. After a preliminary translation, we may assume that $A_j \neq 0$ for $j = 1, \dots, n-1$. Choose a point $A_n^* > 0$ on the real line, and consider the fractional linear map defined by

$$\Phi(z) = A_n^* - \frac{1}{z}.$$

Then Φ is an automorphism of the upper half-plane. Let $A_k^* = \Phi(A_k)$ for $k = 1, \dots, n-1$, and note that $A_n^* = \Phi(\infty)$. Then

$$(F \circ \Phi^{-1})(A_k^*) = a_k \quad \text{for all } k = 1, 2, \dots, n.$$

We can now apply the Schwarz-Christoffel formula just proved to find that

$$(F \circ \Phi^{-1})(z') = C_1 \int_0^{z'} \frac{d\zeta}{(\zeta - A_1^*)^{\beta_1} \cdots (\zeta - A_n^*)^{\beta_n}} + C_2.$$

The change of variables $\zeta = \Phi(w)$ satisfies $d\zeta = dw/w^2$, and since we can write $2 = \beta_1 + \cdots + \beta_n$, we obtain

$$\begin{aligned} (F \circ \Phi^{-1})(z') &= C_1 \int_0^{\Phi^{-1}(z')} \frac{dw}{(w(A_n^* - A_1^*) - 1)^{\beta_1} \cdots (w(A_n^* - A_{n-1}^*) - 1)^{\beta_{n-1}}} + C_2' \\ &= C_1' \int_0^{\Phi^{-1}(z')} \frac{dw}{(w - 1/(A_n^* - A_1^*))^{\beta_1} \cdots (w - 1/(A_n^* - A_{n-1}^*))^{\beta_{n-1}}} + C_2'. \end{aligned}$$

Finally, we note that $1/(A_n^* - A_k^*) = A_k$ and set $\Phi^{-1}(z') = z$ in the above equation to conclude that

$$F(z) = C_1' \int_0^z \frac{dw}{(w - A_1)^{\beta_1} \cdots (w - A_{n-1})^{\beta_{n-1}}} + C_2',$$

as was to be shown.

4.5 Return to elliptic integrals

We consider again the elliptic integral

$$I(z) = \int_0^z \frac{d\zeta}{[(1 - \zeta^2)(1 - k^2\zeta^2)]^{1/2}} \quad \text{with } 0 < k < 1,$$

which arose in Example 3 of Section 4.1. We saw that it mapped the real axis to the rectangle R with vertices $-K$, K , $K + iK'$, and $-K + iK'$. We will now see that this mapping is a conformal mapping of $\mathbb{H}$ to the interior of R .

According to Theorem 4.6 there is a conformal map F to the rectangle, that maps four points on the real axis to the vertices of R . By preceding this map with a suitable automorphism of $\mathbb{H}$ we may assume that F maps $-1, 0, 1$ to $-K, 0, K$, respectively. Indeed, by using a preliminary automorphism, we may assume that $-K, 0, K$ are the images of points $A_1, 0, A_2$ with $A_1 < 0 < A_2$; then we can further take $A_1 = -1$ and $A_2 = 1$. See Exercise 15.

Next, let ℓ be chosen with $0 < \ell < 1$, so that $1/\ell$ is the point on the real line mapped by F to the vertex $K + iK'$, which is the vertex next in order after $-K$ and K . We claim that $F(-1/\ell)$ is the vertex $-K + iK'$. Indeed, if $F^*(z) = -F(-\bar{z})$, then by the symmetry of R , F^* is also a conformal map of $\mathbb{H}$ to R ; moreover $F^*(0) = 0$, and $F^*(\pm 1) = \pm K$. Thus $F^{-1} \circ F^*$ is an automorphism of $\mathbb{H}$ that fixes the points $-1, 0$, and 1 . Hence $F^{-1} \circ F^*$ is the identity (see Exercise 15), and $F = F^*$, from which it follows that

$$F(-1/\ell) = -\overline{F}(1/\ell) = -K + iK'.$$

Therefore, by Theorem 4.6

$$F(z) = c_1 \int_0^z \frac{d\zeta}{[(1 - \zeta^2)(1 - \ell^2 \zeta^2)]^{1/2}} + c_2.$$

Setting $z = 0$ gives $c_2 = 0$, and letting $z = 1, z = 1/\ell$, yields

$$K(k) = c_1 K(\ell) \quad \text{and} \quad K'(k) = c_1 K'(\ell),$$

where

$$K(k) = \int_0^1 \frac{dx}{[(1 - x^2)(1 - k^2 x^2)]^{1/2}},$$

$$K'(k) = \int_1^{1/k} \frac{dx}{[(x^2 - 1)(1 - k^2 x^2)]^{1/2}}.$$

Now $K(k)$ is clearly strictly increasing as k varies in $(0, 1)$. Moreover, a change of variables (Exercise 24) establishes the identity

$$K'(k) = K(\tilde{k}) \quad \text{where } \tilde{k}^2 = 1 - k^2 \text{ and } \tilde{k} > 0,$$

and this shows that $K'(k)$ is strictly decreasing. Hence $K(k)/K'(k)$ is strictly increasing. Since $K(k)/K'(k) = K(\ell)/K'(\ell)$, we must have $k = \ell$, and finally $c_1 = 1$. This shows that $I(z) = F(z)$, and hence I is conformal, as was to be proved.

A final observation is of significance. A basic insight into elliptic integrals is obtained by passing to their inverse functions. We therefore consider $z \mapsto \operatorname{sn}(z)$, the inverse map of $z \mapsto I(z)$.⁹ It transforms the closed rectangle into the closed upper half-plane. Now consider the series of rectangles $R = R_0, R_1, R_2, \dots$ gotten by reflecting successively along the lower sides (Figure 10).

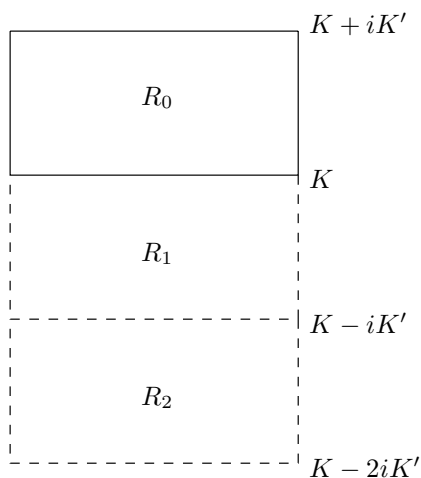


Figure 10. Reflections of $R = R_0$

With $\operatorname{sn}(z)$ defined in R_0 , we can by the reflection principle extend it to R_1 by setting $\operatorname{sn}(z) = \overline{\operatorname{sn}(\bar{z})}$ whenever $z \in R_1$ (note that then $\bar{z} \in R_0$). Next we can extend $\operatorname{sn}(z)$ to R_2 by setting $\operatorname{sn}(z) = \overline{\operatorname{sn}(-iK' + \bar{z})}$ if $z \in R_2$ and noting that if $z \in R_2$, then $-iK' + \bar{z} \in R_1$. Combining these reflections and continuing this way we see that we can extend $\operatorname{sn}(z)$ in the entire strip $-K < \operatorname{Re}(z) < K$, so that $\operatorname{sn}(z) = \operatorname{sn}(z + 2iK')$.

Similarly, by reflecting in a series of horizontal rectangles, and combining these with the previous reflections, we see that $\operatorname{sn}(z)$ can be continued to the complex plane and also satisfies $\operatorname{sn}(z) = \operatorname{sn}(z + 4K)$. Thus $\operatorname{sn}(z)$ is *doubly periodic* (with periods $4K$ and $2iK'$). A further examination

⁹The notation $\operatorname{sn}(z)$ in somewhat different form is due to Jacobi, and was adopted because of the analogy with $\sin z$.

shows that the only singularities $\text{sn}(z)$ are poles. Functions of this type, called “elliptic functions,” are the subject of the next chapter.

5 Exercises

1. A holomorphic mapping $f : U \rightarrow V$ is a **local bijection** on U if for every $z \in U$ there exists an open disc $D \subset U$ centered at z , so that $f : D \rightarrow f(D)$ is a bijection.

Prove that a holomorphic map $f : U \rightarrow V$ is a local bijection on U if and only if $f'(z) \neq 0$ for all $z \in U$.

[Hint: Use Rouché’s theorem as in the proof of Proposition 1.1.]

2. Suppose $F(z)$ is holomorphic near $z = z_0$ and $F(z_0) = F'(z_0) = 0$, while $F''(z_0) \neq 0$. Show that there are two curves Γ_1 and Γ_2 that pass through z_0 , are orthogonal at z_0 , and so that F restricted to Γ_1 is real and has a minimum at z_0 , while F restricted to Γ_2 is also real but has a maximum at z_0 .

[Hint: Write $F(z) = (g(z))^2$ for z near z_0 , and consider the mapping $z \mapsto g(z)$ and its inverse.]

3. Suppose U and V are conformally equivalent. Prove that if U is simply connected, then so is V . Note that this conclusion remains valid if we merely assume that there exists a continuous bijection between U and V .

4. Does there exist a holomorphic surjection from the unit disc to $\mathbb{C}$?

[Hint: Move the upper half-plane “down” and then square it to get $\mathbb{C}$.]

5. Prove that $f(z) = -\frac{1}{2}(z + 1/z)$ is a conformal map from the half-disc $\{z = x + iy : |z| < 1, y > 0\}$ to the upper half-plane.

[Hint: The equation $f(z) = w$ reduces to the quadratic equation $z^2 + 2wz + 1 = 0$, which has two distinct roots in $\mathbb{C}$ whenever $w \neq \pm 1$. This is certainly the case if $w \in \mathbb{H}$.]

6. Give another proof of Lemma 1.3 by showing directly that the Laplacian of $u \circ F$ is zero.

[Hint: The real and imaginary parts of F satisfy the Cauchy-Riemann equations.]

7. Provide all the details in the proof of the formula for the solution of the Dirichlet problem in a strip discussed in Section 1.3. Recall that it suffices to compute the solution at the points $z = iy$ with $0 < y < 1$.

(a) Show that if $re^{i\theta} = G(iy)$, then

$$re^{i\theta} = i \frac{\cos \pi y}{1 + \sin \pi y}.$$

This leads to two separate cases: either $0 < y \leq 1/2$ and $\theta = \pi/2$, or $1/2 \leq$